Bimanual Imitation Learning for Decorative Napkin Folding on Low-Cost Hardware: End-Effector Geometry and the Stage-Composition Bottleneck

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***Abstract*—Decorative napkin folding is a compact but representative deformable-object manipulation task: it is economically relevant in hospitality, yet the highly non-linear dynamics of linen make simulation-based reinforcement learning impractical because of the sim-to-real gap. We present a complete, low-cost bimanual system that learns to fold linen napkins directly from human teleoperation demonstrations, using two 6-DOF SO-101 arms, a three-camera RGB perception stack, and the open-source LeRobot framework. We make three contributions. First, we study end-effector geometry for deformable-cloth grasping and show that a geometry-entrapment "tangential scoop" eliminates post-grasp cloth crumpling (0% crumple across all conditions), whereas force-closure designs crumple the cloth in 47–100% of trials. Second, we compare imitation-learning policy families on a 499-episode, 550k-frame dataset: a 450M-parameter vision-language-action model (SmolVLA) reaches 70% end-to-end fold success (mean cycle time $40.2\,$s, mean edge deviation $6\,$mm), while an Action-Chunking Transformer (ACT) trained on the same full-task data fails entirely ($0/20$). Third, through stage-isolated ablations we localize this failure to a *stage-composition bottleneck* at the pickup-to-fold phase boundary: ACT policies trained on pickup-only or fold-only data each succeed reliably in isolation, but a single monolithic policy cannot resolve the multimodal action distribution in which pickup-like and fold-like actions are both plausible under near-identical observations. We discuss implications for edge-deployable cloth manipulation and outline stage-aware reward modeling and flow-matching policies as paths to closing the remaining gap.**

***Index Terms*—deformable object manipulation, imitation learning, bimanual manipulation, vision-language-action models, cloth folding, low-cost robotics**

I. Introduction

Robotic manipulation of deformable objects remains substantially harder than manipulation of rigid bodies. Cloth in particular occupies a near-infinite configuration space, deforms continuously under contact, self-occludes, and exhibits dynamics that depend on weave, weight, and finish in ways that current physics engines do not reproduce faithfully. As a consequence, the standard recipe for learned manipulation—training a policy in simulation with reinforcement learning (RL) and transferring it to hardware—is largely unavailable for cloth, because the sim-to-real gap is prohibitive [11].

We study this problem through a deliberately narrow but practically grounded task: folding a square linen napkin into a single decorative fold using two low-cost arms (Fig. 1). The task is attractive for three reasons. (i) It is economically meaningful: premium hospitality venues spend one to three hours of paid staff time per day on decorative folding, a repetitive duty that adds no direct revenue. (ii) It isolates the core difficulty of deformable manipulation—reliable, presentation-grade folding of thin cloth—while bounding the perception and workspace problem to a controlled tabletop. (iii) It is achievable on inexpensive, open hardware, which makes the resulting pipeline reproducible.

Because simulation is not a viable training source, we adopt imitation learning (IL) from real human demonstrations. Within IL we evaluate the two dominant contemporary policy families on the same data: Action-Chunking Transformers (ACT) [1], which are prized for data efficiency, and compact vision-language-action (VLA) models such as SmolVLA [3], which inherit semantic priors from large-scale pretraining. A central practical constraint shapes this choice: a deployable system must run inference on an edge accelerator (e.g., an NVIDIA Jetson Orin with $8\,$GB of VRAM), which rules out the largest VLAs and motivates compact architectures.

Our experiments produce a partial-success system and, more usefully, a clear diagnosis of *why* the remaining gap exists. The headline empirical findings are: a SmolVLA policy folds end-to-end in 70% of trials, comfortably inside the cycle-time budget; a same-data ACT policy fails completely; and stage-isolated ablations reveal that the bottleneck is not data volume or training stability but the policy's inability to *compose* two distinct behavioral modes (grasp acquisition and folding) across a sharp phase boundary.

Contributions.

1) A complete, reproducible, low-cost bimanual IL pipeline for deformable napkin folding, built on open hardware (SO-101) and the open-source LeRobot framework [9], including a GPU-side inference path that restores real-

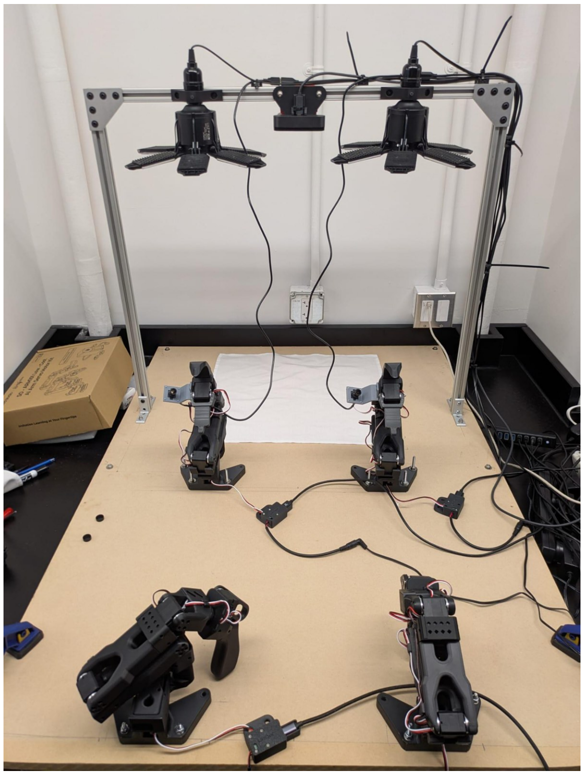

Fig. 1. The integrated platform. Two 6-DOF SO-101 follower arms are mounted to a continuous hardboard base at the outer edges of the workspace; an aluminium-extrusion gantry carries an overhead RGB camera and DC ring lighting, with two additional wrist-mounted cameras. A matched pair of leader arms (used only for data collection) is fixed to the same base, keeping the teleoperation-to-autonomy pipeline aligned without re-calibration.

time (30 Hz) control.

2) A controlled study of end-effector geometry for cloth grasping that quantifies a force-closure vs. geometry-entrapment trade-off and identifies post-grasp crumpling—not pickup speed—as the dominant determinant of downstream fold quality.
3) A head-to-head comparison of ACT and SmolVLA on identical data, and a stage-isolation diagnostic that localizes end-to-end failure to a *stage-composition bottleneck* at the pickup-to-fold boundary, with concrete architectural remedies.

II. Related Work

Deformable cloth manipulation. Robotic cloth folding has a long history; recent surveys catalogue the breadth of perception and control strategies developed for the domain [11]. Classical pipelines combine cloth-state estimation with scripted or primitive-parameterized actions. SpeedFolding [10], for instance, learns to predict gripper-pose pairs for a small set of bimanual primitives and folds garments from a crumpled state in under 120 s at 30–40 folds per hour. Such systems excel at category-level garment folding but rely on engineered primitives and instruction-specified fold lines. We instead learn a continuous visuomotor policy end-to-end, trading some structure for the ability to retarget the same hardware to new folds via re-demonstration rather than re-engineering.

Imitation learning for manipulation. Behavior cloning from teleoperated demonstrations has become a practical alternative to RL for contact-rich tasks where reward design and simulation are difficult. ACT [1] predicts short temporal "chunks" of actions and demonstrates competent bimanual manipulation from only tens of demonstrations, which makes it an attractive starting point for data-limited projects. Diffusion Policy [2] models the full multimodal action distribution rather than a point estimate, a property that is particularly relevant to tasks with branching behavior.

Vision-language-action models. A second line of work couples a pretrained vision-language backbone with an action head, yielding broad generalization and cross-embodiment transfer [5], [6]. These models are typically large; SmolVLA [3] targets affordable robotics with a 450M-parameter backbone ($\sim$4 GB VRAM), and X-VLA [4] explores scalable cross-embodiment soft prompting. Several VLA variants generate actions via flow matching [6], [7], and laundry folding is a recurring evaluation task for these systems—underscoring that deformable folding is now a benchmark frontier for generalist policies. Our setting is the resource-constrained end of this spectrum: we ask how far a compact VLA can be pushed on a single deformable task with a few hundred demonstrations and an edge inference budget.

Stage-aware control. For long-horizon tasks, recent work decomposes behavior into stages and supplies dense, stage-level feedback. Stage-Aware Reward Modeling (SARM) [8] converts a sparse binary task reward into a continuous, differentiable signal at stage boundaries, enabling RL fine-tuning on top of an IL initialization. We connect our negative finding—monolithic policies failing to compose stages—to exactly this class of remedy.

III. System Overview

The system is a dual-arm tabletop platform that learns to fold a manually placed square napkin into a single decorative fold. It comprises a structural frame, two manipulators, a fixed perception stack, and a learned controller.

Hardware. We use two open-source SO-101 6-DOF arms as *follower* manipulators, with custom end-effectors (Section IV). For data collection, a matched pair of SO-101 *leader* arms is teleoperated by a human; the follower arms mirror the leader poses in real time. All four arms are bolted at fixed positions to a single continuous hardboard base, which we found empirically stiff enough to hold the arms square during folding while remaining cheap, portable ($\sim$12 kg total), and easy to modify—a deliberate departure from an earlier optical-breadboard build that was rigid but immobile and expensive.

Workspace geometry. We evaluated two arm-spacing layouts (Fig. 2): center-mounted arms with overlapping reach envelopes, and outer-edge mounting with the fold zone centered between the arms. We selected the outer-edge layout because it reduces inter-arm collision risk during fold-over motions and gives each arm sufficient independent reach to capture a

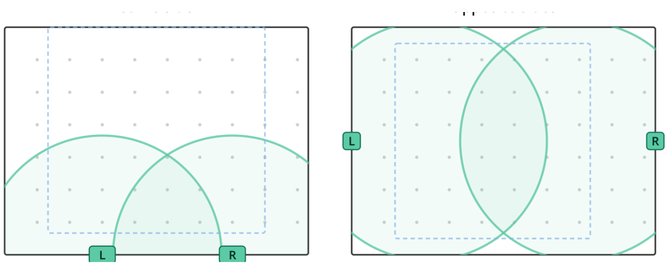

Fig. 2. Arm-mounting geometry trade-off. Left: centered arms with overlapping reach envelopes concentrated near the bases. Right: separated outer-edge layout with the fold zone centered between the arms (selected). The dashed rectangle marks the workspace; shaded regions are per-arm reach.

Fig. 3. Synchronized frames from the three-camera stack during a demonstration: overhead view (left) for global napkin pose and two wrist-mounted views (center, right) for local gripper-to-cloth geometry.

napkin corner; the trade-off is that cooperative grips on a single corner require extended poses, which the folding trajectories are designed to accommodate.

Perception. Three RGB cameras (640×480, 30 fps, USB) observe the workspace: one overhead camera fixed to the gantry, plus two wrist-mounted cameras (Fig. 3). The overhead view captures global napkin pose; the wrist views capture local gripper-to-cloth geometry. This redundancy reduces the chance that a critical feature is occluded in all views simultaneously. To limit distribution shift between demonstration and deployment, camera viewpoints, lighting, and arm-base positions are held fixed across sessions, and the operator places each napkin within a standardized workspace region.

Software stack and real-time inference. The pipeline is built on the open-source LeRobot framework [9], which provides integrated data collection, dataset management, training, and inference (Fig. 4). LeRobot's default inference path performs image preprocessing (resize, normalize, color-space conversion) on the CPU before handing observations to the GPU. On our initial workstation this created a CPU bottleneck that starved the GPU and depressed the control-loop frequency; at low frequencies the gap between observation and action grows and motion becomes jerky and unresponsive. We replaced this with a custom inference path that passes raw camera frames directly to the GPU, where the policy's own vision encoder performs preprocessing within the forward pass. This eliminated the CPU bottleneck and restored the target 30 Hz control loop. Algorithm 1 and Fig. 5 summarize the resulting closed loop.

Performance targets. The platform is benchmarked against operationally derived requirements (summarized in Table I): a cycle time of ≤ 90 s per napkin (human folding ranges 7–40 s);

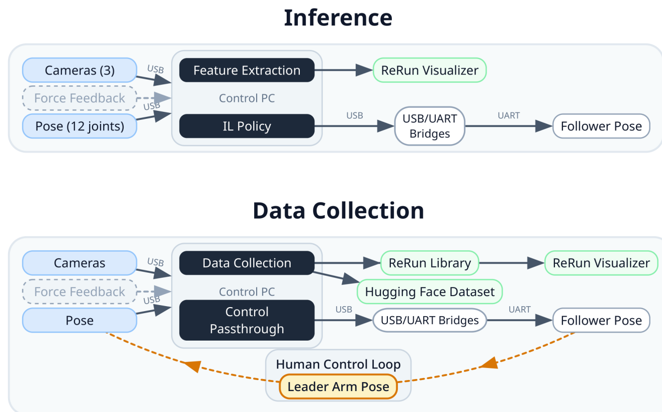

Fig. 4. System dataflow. Top: inference path—cameras and joint-angle pose feed the control PC, which runs the learned policy and drives the follower arms through USB–UART bridges. Bottom: data-collection path—a human-in-the-loop teleoperates the leader arms while synchronized frames and poses are logged to a dataset.

Algorithm 1 Real-time bimanual control loop (30 Hz)

Input: trained policy π, chunk length H, control period $\Delta t = 1/30$ s
loop
  $o \leftarrow$ GETOBSERVATION() $\triangleright$ 12 joint angles + 3 RGB frames
  $\hat{a}_{t:t+H} \leftarrow \pi(o)$ $\triangleright$ GPU-side preprocessing + inference
  $a_t \leftarrow$ TEMPORALENSEMBLE($\hat{a}$, α=0.01) $\triangleright$ n_{steps}=1
  SENDACTION(a_t) $\triangleright$ via USB–UART bridge
  PRECISESLEEP to hold Δt
end loop

a final edge-alignment error of $\leq$ 10 mm for "table-ready" fine-dining presentation; a footprint of $\leq$ 36×36 in to fit a standard prep table; and a target first-attempt success rate of $\geq$ 80% (16/20) for a commercially viable unit.

IV. End-Effector Design

The end-effector is the only physical interface between robot and cloth, and its geometry sets pickup reliability, the amount of cloth deformation induced, and therefore the difficulty of every downstream control step. We treat its design as a controlled experiment.

Design space. A single-DOF rotary gripper can engage a flat cloth edge in two qualitatively different ways. *Force-closure* designs clamp the cloth between opposing faces, holding it by friction and compression. *Geometry-entrapment* designs slide a curved surface beneath the cloth edge and lift before closing, capturing the cloth without compressive loading. We fabricated four prototypes (all 3D-printed in PLA+) that bracket this space: three force-closure variants (OG narrow-tip pinch, EEJ thick-jaw pinch, EED inverse close with wide parallel pads) and one geometry-entrapment variant (EES tangential scoop).

Protocol. Each design was evaluated through teleoperated pickup trials on both flat and pre-folded napkins, approached from both the first-encounter (F.E.) and second-encounter

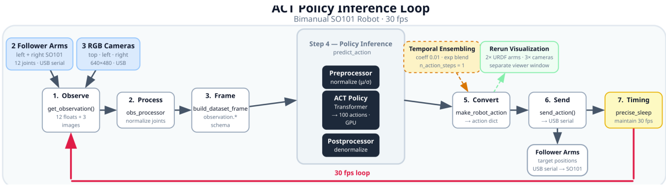

Fig. 5. The seven-step inference loop running at 30 Hz: observe, preprocess, frame, policy inference (with temporal ensembling), convert, send, and precise-sleep timing. The ACT variant emits a 100-action chunk per step.

TABLE I
KEY PERFORMANCE REQUIREMENTS AND THEIR JUSTIFICATION.

Requirement	Target	Justification
Cycle time	≤ 90 s	Human benchmark 7–40 s; batch processing
Fold accuracy	≤ 10 mm	"Table-ready" presentation standard
First-attempt success	$\geq 80\%$	Minimize rework before deployment
Footprint	$\leq 36{\times}36$ in	Standard commercial prep table
Supply voltage	120 V AC	North-American venue infrastructure
Emergency stop	≤ 0.5 s	ISO 10218-1; ISO/TS 15066

(S.E.) sides, to span the grasp conditions the learned policy would later face. We recorded mean pickup time and *crumple rate*—the fraction of trials that produced visible cloth deformation after grasp.

Results. Table II reports the outcome. The decisive metric is crumple rate, not speed. The geometry-entrapment EES scoop achieved a 0% crumple rate across *all* conditions, while every force-closure design crumpled the cloth in at least 47% of trials, and two of them (OG, EED) in 100%. EES pays a modest 1–2 s pickup penalty relative to the fastest design (OG), but that penalty is negligible against the 90 s cycle budget. Crucially, post-grasp crumpling propagates downstream: a crumpled initial grasp produces corner misalignment and crease defects that the folding policy cannot subsequently correct. We therefore selected EES for integration. The result confirms the design hypothesis that geometry-entrapment is fundamentally better suited to thin deformable linen than force-closure, because it eliminates compressive cloth loading by construction.

V. Learning Approach

A. Data Collection

Demonstrations were gathered by kinesthetic teleoperation: a human operator manipulates the SO-101 leader arms, the follower arms mirror the leader poses, and the system records synchronized 12-dimensional joint-angle trajectories together with RGB streams from all three cameras at 30 fps. Each complete pickup-and-fold sequence is one episode. We collected two categories:

- **Full-task** (400 episodes, $\approx$ 4 h): pick up a flat napkin, align the corners, and execute the complete fold.

TABLE II
END-EFFECTOR COMPARISON. PICKUP TIME t IN SECONDS (LOWER IS FASTER); CRUMPLE RATE c IN PERCENT (LOWER IS BETTER). FC: FORCE-CLOSURE; GE: GEOMETRY-ENTRAPMENT.

		Pickup time t (s)		Crumple c (%)	
Design	Type	flat	folded	flat	folded
OG	FC	**4.03**	**3.60**	100	100
EEJ	FC	12.68	9.44	77	47
EED	FC	5.27	4.17	100	100
EES	GE	5.74	4.57	**0**	**0**

- **Pickup-only** (99 episodes, $\approx$ 1 h): perform only the pickup and corner-capture phase. These were added after early ACT runs revealed pickup as the dominant failure mode, to increase the density of pickup-state coverage.

In total the corpus comprises 499 episodes and over 550k synchronized frames. All data is saved locally, backed up to an institutional HPC cluster, and released publicly to support reproducibility.

B. Policy Families

We evaluated three families, constrained throughout by the edge inference budget (a deployable policy must fit in $\sim$8 GB of VRAM).

ACT. ACT encodes short temporal chunks of demonstrated actions and decodes a chunk of 100 future joint-angle targets per inference step [1]. Its main appeal is data efficiency and a small memory footprint. We augmented the baseline with two modifications: *temporal ensembling* at inference (exponentially blending overlapping chunks, coefficient 0.01, $n_{\text{steps}}{=}1$) to smooth jitter, and *cosine-annealed learning-rate scheduling* to stabilize convergence. Both improved trajectory smoothness and training reliability relative to the baseline.

VLA models. We evaluated two VLAs. An early X-VLA [4] experiment produced poor results—the policy commanded trajectories that drove the arms into the table, indicating an action-space calibration mismatch—and was not pursued further under time constraints. SmolVLA [3] (450M parameters, $\sim$4 GB VRAM) trained successfully and produced the most performant policy in our evaluation. It was trained *without* the temporal-ensembling and learning-rate refinements applied to ACT, as adapting those techniques to the VLA training loop was out of scope; flow matching [7] would allow them to transfer naturally in future work.

Diffusion Policy. Diffusion-based action generation [2] models the full multimodal action distribution and is a strong candidate for the branching pickup-to-fold transition. It remained future work for this study.

C. Training

ACT was trained for 100k gradient steps ($\approx$ 15 h on a local consumer GPU). The training loss decreased smoothly and monotonically with no divergence (Fig. 6); while the loss is not a direct measure of task success, the clean curve gives confidence that the data pipeline and training script are

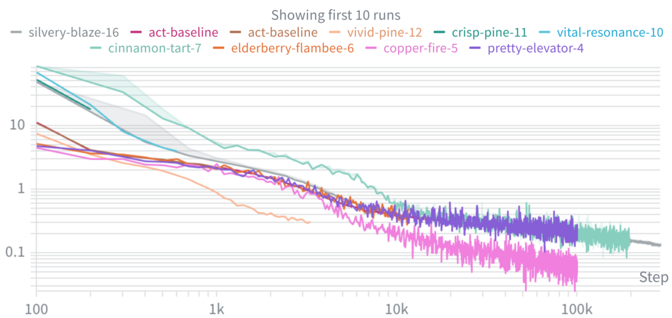

Fig. 6. ACT training loss over 100k gradient steps (log–log). The smooth, monotonically decreasing curve indicates a healthy data and training pipeline—yet, as Section VI shows, a converged loss is fully consistent with 0% end-to-end task success when the failure is one of stage composition rather than representation.

TABLE III
TRAINING ENVIRONMENTS AND THEIR PRACTICAL TRADE-OFFS.

Environment	GPU	Notes
Local (initial)	RTX 2070	Also used for inference; CPU starvation motivated the upgrade
Local (upgraded)	RTX 4070	Primary workstation; fastest iteration cycle
Cloud	H100 / RTX 5090	Highest throughput; data-transfer overhead
Institutional HPC	A40	Queue latency / staging hurt rapid iteration

free of critical errors. Extending to 200k steps produced no further meaningful reduction, suggesting ACT reached the representational capacity of the architecture given this dataset. SmolVLA was trained on the full-task dataset only.

A dataset-merging limitation. We were unable to merge the pickup-only episodes into SmolVLA training: differences in episode metadata and frame indexing between the two collection runs caused intermittent dataloader crashes when sampling across the partition boundary. SmolVLA was therefore trained on the 400 full-task episodes alone. Resolving this schema incompatibility is a prerequisite for combined-corpus training and is, we believe, the single highest-leverage fix available (Section VII).

Training infrastructure. We trained across local consumer GPUs, a cloud provider, and an institutional HPC cluster (Table III). Counter-intuitively, local training had the fastest end-to-end iteration cycle: the overhead of staging the 550k-frame image dataset and waiting in HPC job queues negated the raw compute advantage of cloud/HPC for short experiments. Cloud GPUs were reserved for long runs (including SmolVLA), where per-step speedups justified the transfer cost.

VI. Experiments and Results

A. Twenty-Cycle Bench Test

The primary verification test mirrors the deployment requirement: complete a fold on a freshly ironed $20'' \times 20''$ linen napkin within 90 s with final edge-alignment error < 10 mm, in $\geq 80\%$ of trials (16/20). For each of 20 trials a napkin was placed flat in a randomized workspace position (no automated feed), the operator issued a start command, and the system ran its full autonomous pipeline. We recorded wall-clock cycle time (stopwatch) and maximum edge deviation against a reference template (calipers, ± 1 mm). A trial counted as a success only if both thresholds were met. The protocol was repeated identically for each trained policy.

TABLE IV
TWENTY-CYCLE BENCH-TEST RESULTS BY POLICY.

Policy	Succ.	Rate	Cycle (s)	Edge (mm)	Notes
ACT (full-task)	0/20	0%	—	—	no grasp
SmolVLA	14/20	70%	40.2	6	best

Results. Table IV summarizes the outcome. SmolVLA was the best policy at 70% success, with a mean cycle time of 40.2 s (well inside the 90 s budget) and mean edge deviation of 6 mm (inside the 10 mm accuracy threshold). Its pretrained vision-language backbone provides observation representations rich enough to partially disambiguate the pickup-to-fold transition. Of SmolVLA's six failures, five traced to unsuccessful initial grasps and one to a mid-fold slip during the crease phase—confirming pickup as the residual bottleneck even for the strongest policy. The same-data ACT policy failed every trial (0/20), typically never establishing a stable grasp and therefore never initiating the fold. No policy reached the 80% deployment threshold; the gap from SmolVLA's 70% to 80% is the central remaining technical milestone.

B. Stage-Isolation Diagnostics

To explain ACT's total failure, we trained stage-isolated variants and tested them in isolation:

- **ACT pickup-only** (99 episodes): achieved reliable grasps in nearly all trials.
- **ACT fold-only** (given a human-assisted, pre-grasped napkin): completed folds reliably, with consistent corner alignment and a well-defined diagonal crease (Fig. 7).
- **ACT full-task** (400 episodes): failed to compose the two stages, either missing the grasp or transitioning erratically between pickup-like and fold-like motions.

The diagnostic is unambiguous: each sub-behavior is individually learnable from this data, but the *combined* behavior is not, for a single monolithic ACT policy. This is a stage-composition problem, not a data-quality or convergence problem.

VII. Analysis and Discussion

Why monolithic ACT fails to compose. The pickup and fold phases occupy distinct regions of the joint observation–action space, but they are connected by a sharp boundary at which the same—or very similar—observation can demand qualitatively different actions (continue grasping vs. begin folding). ACT predicts a single action chunk per timestep and cannot represent this multimodality: when both pickup-like and fold-like actions are plausible under a near-identical

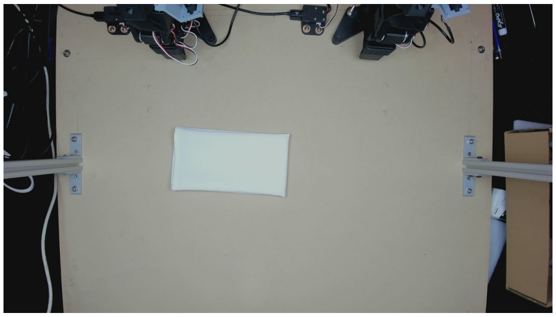

Fig. 7. Completed fold produced by the fold-only policy given a stable grasp: a clean diagonal crease with aligned corners and a well-defined edge. When the pickup bottleneck is bypassed, fold quality meets the presentation standard—the system's ceiling is set by grasp acquisition and stage composition, not by the folding motion.

observation, the policy must collapse them, and the result is the erratic boundary behavior we observe. SmolVLA's pretrained encoder yields richer observation features that partially disambiguate the transition, which explains its superior end-to-end performance—but its 70% ceiling shows that better representations alone do not fully resolve stage composition.

The loss-curve trap. ACT's training loss plateaued smoothly without divergence (Fig. 6), which could be mistaken for a healthy, near-optimal model. Yet the full-task policy succeeds on 0% of trials. The lesson is methodological: behavior-cloning loss measures prediction error on demonstrated chunks, not task success, and a converged loss curve is fully consistent with catastrophic end-to-end failure when the failure is one of composition rather than representation. Task-level evaluation, not loss, must gate progress.

Fold quality is not the limiter. When the pickup bottleneck is bypassed (fold-only policy or human-assisted grasp), the resulting fold is acceptable: a clean diagonal crease with aligned corners (Fig. 7). The system's quality ceiling is set by grasp acquisition and stage composition, not by the folding motion itself. We note that fold quality is currently scored only as binary success/failure; a continuous, vision-based fold-quality metric is needed to provide gradient information for both evaluation and future RL fine-tuning.

Paths to closing the gap. Three concrete directions follow directly from the analysis:

1) *Stage-aware decomposition and RL fine-tuning.* SARM [8] explicitly decomposes the task into stages (pickup, align, fold, release) and trains a per-stage reward model, with a meta-controller sequencing stages. This converts the sparse binary task reward into a dense, differentiable signal—a prerequisite for RL fine-tuning on top of an IL initialization, and a principled remedy for the composition failure we observe.
2) *Dataset merging and combined training.* Resolving the LeRobot schema incompatibility would let SmolVLA train on the full 499-episode corpus, adding the pickup coverage that addresses its dominant residual failure mode.
3) *Generative action heads.* Diffusion Policy [2] and flow matching [6], [7] natively represent multimodal action distributions and are therefore well-matched to the branching pickup-to-fold transition; flow matching in particular would let the temporal-ensembling and learning-rate refinements validated on ACT transfer to a generative VLA.

Operational validation of the task. Separate from the technical evaluation, we ran a week-long concierge study at a working restaurant in which a staff member manually produced the napkin output the system is intended to automate. Folding consumed ≈ 1.9 h/day (consistent with our 1–3 h estimate), its removal let staff close 30–60 min earlier, napkin usage tracked guest count at 1.08 napkins/guest, and staff sentiment was uniformly positive. A parallel customer-discovery campaign across 19 venues yielded a 22% letter-of-intent conversion. These results confirm the problem is real and the product direction is sound; the remaining gate is purely the technical 70% $\rightarrow$ 80% fold-success closure. (Selection bias from a single affiliated venue is a known limitation; a second unaffiliated trial is planned.)

Intellectual-property landscape. A prior-art survey across dual-arm deformable control, airflow-assisted textile handling, household garment folding, and garment-manufacturing fabric handling found no patent covering the specific integrated architecture used here (staged dual-arm fold execution, perception tuned for presentation-grade napkin geometry, and cloth-specific end-effectors). The software stack rests on permissively licensed open-source components (LeRobot, SmolVLA weights, PyTorch), none carrying copyleft obligations, so a closed-source commercial release is license-compatible.

Limitations. The evaluation is a single-fold, single-napkin-geometry, controlled-lighting study with a manually placed napkin, and all 499 episodes were collected by a single operator, which may embed operator-specific motion habits and limit generalization. The system is not yet at the 80% deployment threshold, and extended-duty stability (thermal drift, power-supply behavior over 100^{+}-napkin runs) has not been formally characterized. These bound the generality of the claims but not the central diagnostic finding, which concerns the structure of the learning problem rather than the specific deployment.

VIII. Conclusion

We presented a low-cost bimanual imitation-learning system for decorative napkin folding and used it to study two questions of broader interest in deformable manipulation. First, end-effector geometry matters more through induced cloth deformation than through speed: a geometry-entrapment scoop eliminated post-grasp crumpling entirely, whereas force-closure designs crumpled cloth in nearly every trial. Second, on identical data a compact VLA (SmolVLA) reached 70% end-to-end success while a monolithic ACT policy failed completely—and stage-isolated ablations localized that failure to a stage-composition bottleneck at the pickup-to-fold boundary, not to data quality or training stability. The practical implication is that, for low-horizon deformable tasks with sharp behavioral phase boundaries, progress depends less on additional compute and more on representations and training signals that handle stage composition explicitly. Stage-aware reward modeling, combined-corpus training, and generative action heads are the natural next steps toward robust, edge-deployable cloth folding.

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